

Yael Konforti

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Hebrew ♦ English ♦ Spanish

RESEARCH PROFILE

PhD researcher at the University of Cambridge investigating algorithmic bias mitigation in AI systems using explainable AI (XAI) techniques, with a strong background in developing and deploying real-time deep learning frameworks on edge hardware. Research interests span trustworthy AI, XAI, representation learning and probabilistic model interpretation.

EDUCATION

Ph.D. in Computer Science and Technology

University of Cambridge, UK

Oct 2024 – Present

Thesis focus: *Algorithmic Bias Mitigation using XAI methods for AI Applications.*

Supervisor: Prof. Mateja Jamnik.

MSc in Industrial Engineering and Management (Applied Statistics Spec.)– Distinction

Ben-Gurion University, Israel | “Meitar” excellence fast track

Oct 2017 – Oct 2019

Thesis: *Probabilistic Interpretation and Visualization of Deep Neural Networks.*

Supervisors: Dr. Aharon Bar Hillel, and Prof. Boaz Lerner.

GPA: 96/100

BSc in Industrial Engineering and Management– Magna Cum Laude (Top 5%)

Ben-Gurion University, Israel

Oct 2014 – Aug 2018

GPA: 88/100

PUBLICATIONS

- Konforti, Y., et. al. (2022). SIGN: Statistical Inference Graphs based on probabilistic Network activity interpretation. IEEE Transactions on Pattern Analysis and Machine Intelligence.
- Konforti, Y., et. al. (2020). Inference graphs for CNN interpretation. In European Conference on Computer Vision (pp. 69-84). Springer, Cham.
- Konforti, Y., et. al. (2026) CB-SLICE: Concept-Based Interpretable Error Slice Discovery. Proceedings of the 43rd International Conference on Machine Learning (ICML 2026), to appear.

INDUSTRY EXPERIENCE

Research Scientist - SightX AI

Oct 2020 – Present

AI-based aerial real-time object detection system, defense sector | Tel-Aviv, Israel

- Designing and deploying deep learning models optimised for real-time small object detection and tracking on in infrared imagery, with the focus on edge device optimisation and efficiency.
- Managing the research group’s codebase, introducing protocols for feature integration, pull-requests and version control — improving team-wide reproducibility.
- Designing the company’s evaluation framework and test-set protocol, enabling rigorous, standardised model benchmarking across deployment scenarios.

Machine Learning Researcher - BioMedSU
Tel-Aviv, Israel

May 2020 – Oct 2020

- Conducting R&D on Variational Autoencoders (VAE) architectures to predict the impact of drug composition on RNA sequencing data.

TEACHING & SUPERVISION EXPERIENCE

MPhil Student Supervision (Co-supervisor) *2025 – Present*
Department of Computer Science, University of Cambridge

- Mentoring an MPhil dissertation on mitigating spurious correlations in AI models; guiding research methodology, experimental design, and academic writing.

Teaching Assistant, Explainable AI (Part III/M.Phil) *Jan 2025 – Present*
Department of Computer Science, University of Cambridge

- Conducting lab demonstrations, guiding students in practical applications of XAI.
- Assessing coursework and assignments.

Teaching Assistant, Production Planning and Control 2 *Mar 2017 – Oct 2019*
Department of Industrial Engineering and Management, Ben-Gurion University

- Conducting weekly practices for a 40 student class.
- Developing course materials, including lectures, projects, and assessments for undergraduate and postgraduate level courses.

CONFERENCE PRESENTATIONS

The 1st Conference of International Israel Data Science Initiatives, Haifa, Israel *2022*

The 18th Israel Conference of Computer Vision, Tel-Aviv, Israel *2020*

The 16th European Conference on Computer Vision, Glasgow, Scotland *2020*

AWARDS AND SCHOLARSHIPS

Faculty Excellence Scholarship, *2017 – 2019*
IEM Department, Ben Gurion University.
Full tuition and life stipend scholarship.

Excellence Scholarship for Women in STEM Fields, *2018*
The Israeli Ministry of Science and Technology.

Outstanding Instructor Award, *2018*
IEM Department, Ben-Gurion University

ACADEMIC SERVICE

Member, Women@CL Initiative, University of Cambridge *2025 – Present*

Organiser, 11th & 12th Conference of Oxbridge Women in Computer Science,
University of Cambridge *2025, 2026*